

Innovation-Safe Compression of Financial Strategy Libraries: Semantics, Complexity, and Algorithms

David Ramirez^{1*}

^{1*}Independent Researcher, Orlando, FL, USA; [ORCID: 0009-0000-3128-5123](#).

Corresponding author(s). E-mail(s): ramirezdavv@gmail.com;

Abstract

Financial strategy libraries impose maintenance, validation, monitoring, governance, computing, and attention burdens. An entry that is redundant for current operation may nevertheless carry a capability used by a later strategy-generation process. We formulate source-relative minimum-burden retention that preserves both a regime-indexed operating frontier and a declared generative closure. Under explicit library-invariance assumptions, these objects form the library component of a controlled finite-state representation; this is a conditional semantic result, not a general minimality claim. Under identity closure, safe retention is exactly weighted set cover on disjoint frontier and module tags. This established combinatorial class yields a restricted NP-completeness boundary and supports enumeration, requirement-mask dynamic programming, exact preprocessing, mixed-integer optimization, weighted greedy construction, and certified deletion. A parameterized family proves that heaviest-safe-first deletion has no constant approximation guarantee. A registered and audited synthetic benchmark verifies exact-method agreement where applicable, reports certified burden gaps, and shows that preprocessing frequently resolves the generated instances; its cold-start concurrent timings are not used to rank algorithms. A retrospective point-in-time portfolio-library study finds large burden-index reductions and many additional innovation options, but its protected common-equity policy never exercises those options and its single ETF replacement loses held-out value. A separately design-locked synthetic mechanism experiment shows why option preservation and option exercise differ: the learner does not recover a small positive bridge value but does recover a prospectively powered one. These are bounded structural, synthetic, and retrospective findings, not evidence of alpha, forecasting ability, causal effects, or deployable performance.

Keywords: financial strategy libraries, innovation-safe compression, weighted set cover, exact algorithms, model governance

1 Introduction

Financial research is cumulative. Candidate signals, transformations, execution rules, and risk controls enter maintained catalogs after development and validation, where they remain available for operation, comparison, and reuse. As a catalog grows, so do its computing footprint, validation and monitoring queue, documentation burden, and the attention needed to understand it. The decision studied here is removal from an *active maintenance and validation set*. It is not destruction of code, data, model history, or records whose archival retention is required by law, contract, audit policy, or disaster-recovery practice.

The apparent redundancy of a strategy depends on which role is being assessed. For current operation, a strategy is redundant if removing it leaves every regime-dependent operating opportunity unchanged. For future research, the same strategy may be the only declared carrier of a signal, transformation, execution component, or risk-control tag used by a later project. A strategy can consequently be dominated at every current belief and still be generatively indispensable. Pruning only the current payoff envelope can preserve what the library can operate now while silently changing what it can help create later. This carrier interpretation is a modeling declaration: in the financial cases the tags describe a fixed tuple grammar, not evidence that the underlying code, data, tests, or organizational knowledge exist only in a strategy row.

We represent these roles by two source-library objects. The operating frontier F_L records the best current operating value attainable from a source library L at each declared belief. The generative closure C_L records the modules available to the modeled strategy-generation process after its closure rules have been applied. Conditional on the paper's finite library-invariance assumptions, the pair (F_L, C_L) is the library component of a controlled representation: replacing L by a sublibrary with the same frontier and closure leaves the declared operating opportunities and generative menus unchanged. The theorem is a factorization result for a specified information architecture, not a generic coarsest-state or minimality theorem. The filtered belief and any active-project variables remain part of the full controlled state.

This semantic reduction defines *innovation-safe compression*. Given a source library, additive retention burdens, and a mandatory inactive strategy, we seek a minimum-burden source sublibrary whose frontier and closure equal those of the source exactly. Equality is source-relative and exact for the declared finite tables: it is not a prediction criterion, statistical certainty, or a claim of equivalence in unobserved markets. The source itself is feasible, but choosing the least-burden feasible representation is a global combinatorial optimization problem. Sequential deletion can certify that every accepted step is safe, yet an inclusion-wise irreducible endpoint need not have minimum burden.

The identity-closure subclass exposes the combinatorial structure sharply. In that subclass, the requirements are tagged to keep frontier-attainment rows distinct from module-carrier rows, including frontier ties and requirements already met by the inactive strategy. A sublibrary is safe if and only if its strategies cover every remaining tagged requirement. This equivalence reduces the semantic retention problem to a weighted covering formulation; it does not claim set cover itself as a new problem. The restricted decision problem is NP-complete, and the broader compression model is NP-hard because it contains that subclass. The reduction therefore supplies both a

modeling bridge to the classical covering literature (Karp 1972; Chvátal 1979) and a reason not to expect local redundancy checks alone to solve global resource allocation.

We address this boundary with a portfolio of algorithms rather than a single method. Complete enumeration provides a small-library oracle. Requirement-mask dynamic programming is exact when the tagged requirement dimension is manageable. Proven preprocessing rules force unique carriers, remove safely dominated or empty variables, merge duplicate coverage and requirement patterns with explicit reconstruction maps, and can sometimes resolve an instance completely. An auditable mixed-integer formulation scales the covering model while retaining solver diagnostics and an independent exact postcheck of the returned library. Weighted greedy construction inherits its classical covering guarantee under identity closure, and reverse deletion can reduce its burden without weakening that guarantee. Deterministic and multi-start deletion heuristics provide certified feasible endpoints, but their feasibility certificates are not certificates of optimality.

The computational evidence asks a narrower question: how do certified burdens differ within one registered synthetic design? Small instances support exact cross-method comparisons, and the adversarial family connects finite calculation to the worst-case theorem. The structured generator frequently collapses under forced preprocessing and contains infeasible requested factor cells. Moreover, recorded wall times include first-call Julia compilation under concurrent worker contention. We therefore use the study for exact agreement, burden quality, failure accounting, and preprocessing prevalence, not for steady-state runtime rankings, complete factorial inference, or a population claim about financial libraries.

The financial evidence is a retrospective point-in-time portfolio-library study over annual common-equity origins with a separately registered ETF replication. It fixes point-in-time membership, delisting treatment, portfolio grammar, compression libraries, and proposal-year policies before opening each held-out evaluation block. The safe libraries retain the registered frontier and all capability artifacts at substantially lower burden indices and expose many more proposal options than budget-matched frontier-only libraries. Yet the protected policy never exercises an added common-equity option, and its single ETF replacement loses held-out value. A separate calibrated synthetic experiment isolates the bridge mechanism under equal current frontier and burden: small positive option value remains hard to detect, whereas a prospectively powered value is recovered. This pairing keeps option creation, statistical exercise, and realized financial value distinct. Neither study establishes population savings, causality, forecasting skill, alpha, or deployable performance.

1.1 Research questions

The analysis is organized around four questions.

- RQ1.** Under which declared library-invariance conditions does frontier-closure equality preserve both current operating opportunities and the modeled strategy-generation menu?

- RQ2.** How does exact minimum-burden preservation become a covering problem under identity closure, what computational barrier follows, and how far can a certified local-deletion rule be from the global optimum?
- RQ3.** Which exact, approximation-guaranteed, mixed-integer, and certified heuristic methods can exploit this structure while preserving exact feasibility and an auditable reconstruction to original strategy identities?
- RQ4.** Within the registered synthetic designs and the retrospective point-in-time financial panel, when does preserving closure create additional options, when does a protected policy exercise them, and what held-out value is realized when it does?

1.2 Contributions

The paper makes four principal contributions.

1. *Conditional semantic representation.* We identify operationally checkable library-invariance conditions under which the operating frontier and generative closure form a sufficient library summary, and we formulate innovation-safe compression as source-relative preservation of both objects. This separates operational from generative redundancy without claiming a generic minimal state.
2. *Complexity and worst-case theory.* For identity closure, we prove the tagged-cover equivalence and NP-completeness of a restricted safe-compression decision problem. A parameterized family gives the exact burden ratio of the heaviest-safe-first endpoint to the global optimum and shows that this named deletion rule has no constant approximation guarantee. The result does not apply to every deletion order.
3. *Certified computational reduction.* We map the semantic constraint to established weighted-cover machinery under identity closure and provide a common exact representation, optimum-preserving preprocessing with reconstruction, enumeration, requirement-mask dynamic programming, an auditable mixed-integer workflow, weighted greedy, and certified deletion. These are implementation and certification contributions, not claims that the underlying generic cover algorithms are new.
4. *Bounded computational and financial evidence.* The locked benchmark supplies exact-agreement, certified burden-quality, preprocessing, and adversarial records within its realized design. The point-in-time portfolio-library study reports exact structural preservation, burden-index compression, option coverage, protected proposal decisions, and held-out contrasts while preserving the licensed-data boundary. The calibrated closure-option experiment then isolates the theory’s bridge mechanism under prespecified null, adverse, low-dose, and powered regimes. Neither layer is generalized beyond its registered instances.

The theoretical, computational, solver, and financial evidence levels are not interchangeable. Universal claims rest on their stated human proofs and, where the theorem ledger says so, narrower kernel-checked declarations. Exact finite computation establishes facts about specified fixtures or instances. Solver diagnostics document a numerical global-search process, whereas independent exact rechecks certify the feasibility and burden of the returned sublibrary. Registered synthetic results describe the locked generators, and retrospective financial results describe the audited source instances. The algorithmic guarantees in this article apply to identity closure unless a

result explicitly states otherwise; we do not claim a covering algorithm for arbitrary nonidentity closure.

OpenAI ChatGPT and Codex assisted with drafting and editing, software development and review, literature-metadata checking, and repository-based analysis and reproducibility tasks. The author reviewed, revised, and verified all AI-assisted material and remains solely responsible for the work. A full disclosure appears in Statements and Declarations.

1.3 Article organization

Section 2 positions the problem relative to covering optimization, policy compression, partial-information control, and financial research governance. Section 3 defines the financial strategy-library model and establishes the productive-state projection. Section 4 develops exact compression, the tagged covering representation, and the complexity result. Section 5 presents preprocessing, exact and approximate algorithms, mixed-integer optimization, and certified deletion. Section 6 interprets unsafe retention, reports the calibrated closure-option mechanism experiment, and summarizes capacity constraints, building on the local/global gap established in Section 5. Section 7 reports the registered computational study, and Section 8 reports the retrospective point-in-time financial study. Section 9 concludes. Proof details, certificates, extended outputs, registered-study documentation, and reproducibility records are provided in Online Resource 1.

2 Literature and problem positioning

The relevant literatures use similar terms—reduction, pruning, compression, coverage, and libraries—for decisions that preserve different objects. We therefore organize the comparison by research problem and preservation target. Established work in operations research, stochastic control, and combinatorial optimization comes first; recent agent-library and automated-strategy systems are considered afterward as application-level comparators. This ordering is important because the paper’s covering algorithms are inherited from a classical problem class, whereas its modeling contribution begins with the dynamic semantics of a retained financial strategy library.

2.1 Partial-information state reduction and semi-Markov control

Belief-state reduction is foundational in partially observed control. [Smallwood and Sondik \(1973\)](#) formulate finite-horizon control on the belief simplex and establish the familiar piecewise-linear convex value representation; [Sondik \(1978\)](#) develops the discounted infinite-horizon counterpart. Random decision durations do not remove the sufficient-statistic logic: [White \(1976\)](#) gives a finite-horizon, partially observed semi-Markov dynamic program on a sufficient statistic. Financial switching and regime-switching models likewise use filtered information to choose among a fixed menu of operating modes or positions ([Lakner 1998](#); [Li et al. 2015](#); [Dai et al. 2010](#)).

Those results supply the control architecture used here, not a novelty claim. The present reduction concerns a different component of state. Conditional on the filtered belief and any active-project variables, it asks which summary of an endogenous strategy library preserves both current operating choices and the modeled law of future strategy generation. The answer under the declared finite factorization is the operating frontier together with the generative closure. Thus the frontier–closure pair summarizes the *library* component of the productive state; it neither replaces the belief nor proposes a new general sufficient statistic for partially observed or semi-Markov control.

Two standard reductions sharpen the distinction. Incremental pruning removes alpha vectors that are redundant in a fixed partially observable dynamic-programming backup (Cassandra et al. 1997). Its preserved object is the pointwise value representation generated by that backup. A pruned vector does not also act as a parent carrying modules for later policy generation. State aggregation and model minimization instead identify states that are behaviorally equivalent for a specified comparison class (Givan et al. 2003). Their decision unit is a state or model component, not a retained strategy entry. Innovation-safe compression therefore differs from both alpha-vector pruning and state aggregation even though all three use exact equality to justify a smaller representation.

2.2 Policy, action, and control-library compression

The closest established library-selection problem is static control-library design. Dey et al. (2012) select and order compact maneuver or trajectory libraries using submodular-sequence structure, trading library size and search effort against expected task performance. The preserved or optimized object is static task usefulness and search efficiency. In the present problem, a retained entry can additionally alter the set of candidates that can be generated later, so static control coverage does not certify generative safety.

Mutti et al. (2022) provide the closest policy-compression comparison. They construct reward-free representatives of a policy space using coverage of induced state–action distributions and formulate the selection as set cover. Their preserved object is reward-independent distributional representativeness for downstream reinforcement-learning tasks. Here the feasible choices are sublibraries of a named source, and feasibility requires exact equality of a regime- or belief-indexed operating frontier and a parent-conditioned generative closure. Reward-free policy compression can therefore retain a representative for broad downstream usefulness without preserving the same objects as innovation-safe compression.

A directionally different problem expands rather than compresses the available action set. In such models the performance object may be learning or regret in a growing menu. Here generation is conditioned on modules carried by retained parents, and the decision is which existing entries can be deleted without changing either current opportunities or attainable descendants. The present problem is therefore distinct from generative action-set expansion: it concerns source-relative closure-preserving deletion rather than costly action creation.

2.3 Weighted covering and combinatorial representation

Set cover is the classical combinatorial core once a finite requirement universe has been specified. Karp’s set-covering decision problem is among the foundational NP-complete problems (Karp 1972). For positive column costs, Chvátal (1979) gives the standard binary weighted-cover formulation and the minimum-cost-per-newly-covered-requirement greedy rule with its harmonic approximation bound. These are established results. The paper neither introduces weighted set cover nor claims a new generic greedy algorithm.

The contribution is the direction of the reduction. The dynamic semantic model first determines what must be preserved: one tagged class records exact source frontier attainment and another records source modules. Under identity closure, a retained source sublibrary is innovation-safe if and only if it covers this tagged requirement universe, after fixing the mandatory inactive strategy. The tagged-cover equivalence is therefore a computational subclass *induced by* the dynamic frontier-closure model. It is not a semantic interpretation imposed on an arbitrary set-cover instance, and it does not extend automatically to a nonidentity closure with complementarities among raw modules.

Computational benchmarking has a separate methodological literature. Performance profiles compare solvers only after defining a common problem set and success measure (Dolan and Moré 2002); feature-balanced benchmark construction, as in MIPLIB 2017, is designed to avoid overrepresenting a narrow instance family (Gleixner et al. 2021). These principles expose limits of the present locked experiment rather than repairing it after the fact. Its methods have unequal success targets, some infeasible generator cells, and a structured block dominated by complete preprocessing. A future registered benchmark must therefore balance realized residual structure and separate feasibility, quality-qualified, and exact-completion targets before using comparative profiles.

Adjacent covering models optimize different objects. Budgeted maximum coverage and monotone submodular maximization seek the largest attainable value under a cardinality or resource constraint (Nemhauser et al. 1978). Submodular cover minimizes cost to reach the full value of a normalized, monotone, integer-valued submodular function (Wolsey 1982). Adaptive submodularity concerns sequential choices whose outcomes are revealed over time and requires adaptive monotonicity and diminishing returns (Golovin and Krause 2011). Exact innovation-safe compression instead minimizes positive active-strategy burden subject to complete coverage of fixed tagged requirements. The maximum-coverage, submodular-cover, and adaptive guarantees do not transfer unless their distinct assumptions are separately proved.

2.4 Resource-constrained and metareasoning decisions

Operations research has long treated information, search, computation, and research budgets as scarce resources. Weitzman (1979) optimizes costly sequential search among alternatives. Baker et al. (1976) allocate an organizational research-and-development budget across competing activities, and Russell and Wefald (1991) value computational actions by their effect on a subsequent decision. These literatures motivate the economic

importance of research resources without supplying a preservation condition for a maintained strategy library.

The distinction is between choosing what to investigate and choosing which representation to retain. Exact innovation-safe compression is a one-time outer decision that minimizes additive burden while holding the source frontier and source closure fixed. The paper’s capacity- and price-constrained variants relax that equality and ask which productive state is worth buying; they are closer to search, metareasoning, and research-budget allocation. Neither variant turns held-out financial performance into a retention score. This separation also explains why a locally safe deletion certificate answers a feasibility question, whereas global minimum-burden retention answers a resource-allocation question.

2.5 Maintained generative libraries

Library learning and program synthesis establish that reusable components can shape later construction. For example, [Bowers et al. \(2023\)](#) synthesize libraries of reusable abstractions rather than treating candidate programs as independent outputs. Its decision unit is internal program structure, and its preserved object is reusable abstraction within synthesis. The present problem instead selects a sublibrary of whole strategies that must jointly preserve a belief-indexed operating envelope and parent-conditioned generation. Innovation-safe compression is therefore not a substitute for procedural compression, provenance, repair, versioning, or rollback controls.

2.6 Automated financial strategy construction and governance

Automated strategy construction predates contemporary agent systems. [Allen and Karjalainen \(1999\)](#) use genetic search to construct technical trading rules, and [Moody and Saffell \(2001\)](#) learn trading policies by direct reinforcement. These studies optimize or evaluate generated strategies; they do not pose minimum-burden retention of a persistent parent library. Likewise, the model-risk literature studies the consequences of model uncertainty and robust risk measurement ([Cont 2006](#); [Glasserman and Xu 2014](#)). Supervisory guidance documents expectations for model inventories, validation, monitoring, and governance ([Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011](#)). This supports the existence of organizational burdens, not the paper’s optimization model or empirical findings.

The finance literature also gives a direct warning about the interpretation of large rule catalogs. [Sullivan et al. \(1999\)](#) evaluate technical-rule performance while accounting for the universe from which the rules were selected, and [White \(2000\)](#) develops a general reality-check framework for data snooping. Multiple-testing concerns are especially acute in large return- predictor collections ([Harvey et al. 2016](#)), while post-publication evidence shows that reported anomaly returns can attenuate out of sample ([McLean and Pontiff 2016](#)). The article’s ex post maximum over enabled descendants is not a corrected performance test under any of these frameworks. It is used only to witness the mechanical implication of preserving or losing the declared catalog, and no significance or investment claim is attached to it.

An accepted recent financial comparator, ReCAP, maintains regime-adaptive portfolio policies (Pan et al. 2026). Its optimized object is regime-specific portfolio performance, not minimum-burden preservation of a source frontier and generative closure. The present paper begins after a finite generation–verification–admission law has been declared and asks which source entries can be removed without changing those two objects. Its financial audits assess retention burden and semantic preservation, not improved returns, forecasting ability, alpha, or deployable performance.

Claim boundary. The paper claims novelty neither for state, value, policy, action-set, control-library, or internal procedural compression nor for weighted cover, metareasoning, maintained skill libraries, or automated strategy generation. The narrower contribution is that, under the declared finite dynamic model, the operating frontier and generative closure jointly define the object that safe retention must preserve; under identity closure, those semantics induce an exact tagged weighted-cover subclass and source-relative certificates. None of the verified comparisons reviewed here establishes that joint formulation. This is a bounded audit conclusion, not a priority claim over all related library work.

3 Financial strategy-library model and sufficient state

This section states a conditional representation theorem for a declared information architecture. The controller observes a filtered belief, operates from a verified strategy library, and may initiate positive-duration research that generates and verifies a candidate before admission. The theorem does not derive library invariance from finance primitives; it identifies the conclusion that follows when the primitives pass the explicit factorization checks below. The retention decision is made once, before the controlled process is evaluated.

3.1 Hidden state, filtered belief, and operating profiles

Let X be a finite hidden market-state set with at least two elements and let B be a finite nonempty set of filtered beliefs. A belief $b \in B$ has an exact rational interpretation $\mu_b \in \Delta_{\mathbb{Q}}(X)$; the already-filtered belief evolves under an exact rational Markov kernel P_B . The controller observes b , not the hidden state $x \in X$.

The finite strategy catalog is S , with distinguished inactive strategy s_0 . Strategy s has immutable state-contingent operating payoff $u_s : X \rightarrow \mathbb{Q}$ and belief-indexed operating profile

$$j_s(b) := \sum_{x \in X} \mu_b(x) u_s(x), \quad j_{s_0}(b) = 0. \quad (3.1)$$

Thus $j_s(b)$ is the payoff from operating s now; research costs and continuation values are separate objects. The formal projection theorem uses the resulting exact profile table as primitive data, so it does not claim a kernel-verified derivation of that table from a hidden-state model.

Partial information matters to the controlled projection through the filtered belief and its transition law. It is not required by the later tagged-cover equivalence, which needs only a finite index set of operating-profile rows. The financial cases in Section 8

use observed SPY-signal bins as such rows; they do not estimate the hidden-state filter in this subsection.

3.2 Modules, closure, and the raw retained library

Let M be a finite module set. Each strategy carries an immutable set $\text{mods}(s) \subseteq M$, with $\text{mods}(s_0) = \emptyset$. A module is a declared reusable capability tag for later research, not a direct operating payoff or necessarily a separately owned software artifact. The declared closure operator

$$\text{cl} : 2^M \longrightarrow 2^M$$

is extensive, monotone, and idempotent. These properties allow capabilities to be generated from combinations of raw modules; identity closure is the special case $\text{cl}(A) = A$.

A raw retained library is a source subset $L \subseteq S$ containing s_0 . Its operating frontier, raw module inventory, and generative closure are

$$F_L(b) := \max_{s \in L} j_s(b), \quad U_L^{\text{raw}} := \bigcup_{s \in L} \text{mods}(s), \quad (3.2)$$

$$C_L := \text{cl}(U_L^{\text{raw}}). \quad (3.3)$$

Finiteness and mandatory inactivity make the maximum attained and imply $F_L(b) \geq 0$. The raw decision state is (b, L) : continuing earns the frontier payoff, advances the belief once, and leaves the library unchanged.

3.3 Generation, verification, admission, and timing

Let Q be a finite project set. At project q , belief b , and closure C , candidate generation follows an exact law

$$G(q, b, C) \in \Delta_{\mathbb{Q}}(\{\text{none}\} \dot{\cup} S).$$

A generated candidate s passes verification with exact probability $\nu(q, b, C, s)$. The admitted law is derived rather than separately chosen:

$$\Gamma(q, b, C)(s) = G(q, b, C)(s)\nu(q, b, C, s), \quad (3.4)$$

and all failed-generation and failed-verification mass is assigned to **none**. Successful admission adds the verified catalog row, whereas every other outcome leaves the source unchanged:

$$L \oplus \text{none} = L, \quad L \oplus s = L \cup \{s\}. \quad (3.5)$$

Each available project has an initiation cost, an operation/suspension flag, and an integer duration $d_q \geq 1$. Its exact joint completion law couples the intervening belief path with the admitted outcome; the two components need not be independent. The incumbent library remains in force until completion, then (3.5) is applied. Full probability-table definitions, null-outcome normalization, path and outcome marginals, and pushforward identities are assigned to Online Resource 1, Section S1.

3.4 The productive library state

Definition 3.1 (Frontier-closure library state). The productive state of a retained library is

$$K_L := (F_L, C_L). \quad (3.6)$$

It is the *library component* of the controlled state. The filtered belief remains part of the decision state, and an unfinished project also requires its identity and remaining duration.

For $K = (F, C)$, define the local compressed update

$$\text{addK}(K, \text{none}) = K, \quad \text{addK}(K, s) = (b \mapsto \max\{F(b), j_s(b)\}, \text{cl}(C \cup \text{mods}(s))).$$

Extensivity and idempotence of closure give

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (3.7)$$

Assumption 3.2 (Finite library-invariant research process). The sets B, S, M, Q are finite, B and every admissible library are nonempty, and all displayed profiles and probability tables are exact. The inactive strategy has zero profile and no modules; closure is extensive, monotone, and idempotent. Generation and primitive verification depend on the incumbent library only through (q, b, C_L) . Project availability, initiation cost, duration, operation flag, and the joint completion law depend on the library only through (b, K_L) . Each duration is positive, the joint law has the declared belief-path and admitted-outcome marginals, and discount satisfies $0 \leq \beta < 1$. Stationary conclusions additionally assume the recorded raw and compressed Bellman operators are contractions.

Operational factorization audit.

The assumption can be checked table by table. Project availability, cost, duration, operation while active, and the joint completion law must be constant whenever (b, F_L, C_L) is constant; candidate generation and verification must be constant whenever (q, b, C_L) is constant; and admission must update the compressed state through addK . A nontrivial finite class satisfying these conditions consists of module-gated projects whose availability is a predicate of (q, b, C_L) , whose candidate and verification tables are indexed by (q, b, C_L) , and whose duration, cost, and completion tables are indexed by (q, b, F_L, C_L) . Correlation between terminal belief and admission remains unrestricted within each indexed joint table.

The condition is substantive. Let two one-active-strategy libraries $L_a = \{s_0, a\}$ and $L_b = \{s_0, b\}$ have identical zero profiles and no modules. Then $K_{L_a} = K_{L_b}$. If a zero-cost, one-period project with deterministic positive completion value is available only when the raw identity a is retained, their feasible actions and values differ. The projection therefore fails. The same failure occurs if a primitive depends directly on raw cardinality, provenance, owner, or archival status. These variables must either be included in the compressed state or ruled out by the declared information architecture.

Theorem 3.3 (Raw-to-compressed controlled Markov projection). *Under Assumption 3.2, the update in (3.7) induces a normalized transition on realizable states K_L . If $K_L = K_{\bar{L}}$, the two raw libraries have identical project availability, cost, duration, incumbent operating reward blocks, and joint terminal-belief-next-compressed-state laws after projection. Hence (b, K_L) is a controlled semi-Markov projection at embedded decision epochs and, for every finite calendar horizon h ,*

$$V_h^{\text{raw}}(b, L) = \bar{V}_h(b, K_L).$$

Under the additional contraction clause, stationary fixed-point values agree under the same projection and a maximizing stationary compressed selector lifts to a Bellman-optimal raw selector.

Proof sketch. Equation (3.7) makes the next compressed state a deterministic pushforward of the admitted raw update. The library-invariance assumptions make this law, every feasible action, and its payoff block constant on each fiber of $L \mapsto K_L$. Strong induction on calendar horizon is well founded because every research duration is positive; corresponding raw and compressed action values, and then their finite maxima, agree. Bellman intertwining and contraction give the stationary clauses. The complete normalization and projection proof is in Online Resource 1, Sections S1–S2. The result permits belief-admission dependence and makes no generic quotient-minimality claim.

3.5 Retention burden and exact source-relative compression

Let $w_{s_0} = 0$ and give every active strategy an exact rational retention weight $w_s > 0$. The one-time burden is

$$W(L) := \sum_{s \in L} w_s. \tag{3.8}$$

It may index storage, validation, monitoring, active maintenance, retrieval, governance, and human attention. Additivity is a modeling assumption. A weight is not automatically an observed monetary or labor cost, and it is not operating payoff or project cost. Burden is not a component of K_L : equal productive states can have different burdens. In the financial cases, “retained” means retained in the active maintenance and validation set; mandatory archival records, code, data, ownership, and reactivation controls remain outside this binary decision.

For a fixed verified source L , exact source-relative compression chooses an inactive-containing sublibrary $L' \subseteq L$ with $K_{L'} = K_L$ and minimum $W(L')$. By Theorem 3.3, this outer retention decision changes only the raw representation, not the declared productive process. Section 4 defines its feasible family and optimization model. Secondary dynamic comparative statics are not needed for this reduction and are assigned to Online Resource 1, Section S3.

4 Exact compression, covering representation, and complexity

Fix a verified finite source library L . Safety is source-relative: compression may delete source entries but may neither import a replacement nor change the declared profiles, modules, or closure operator.

4.1 Safe feasibility and productive preservation

The safe-feasible family and its global minimum-weight correspondence are

$$\mathcal{P}_{\text{safe}}(L) := \{L' \subseteq L : s_0 \in L', F_{L'} = F_L, C_{L'} = C_L\}, \quad (4.1)$$

$$\mathcal{M}_{\text{safe}}(L) := \arg \min_{L' \in \mathcal{P}_{\text{safe}}(L)} W(L'). \quad (4.2)$$

Equality is exact at every declared belief and for the entire declared generative closure. In particular, a currently dominated strategy is not safe to remove merely because it contributes no current frontier value.

Theorem 4.1 (Minimum-resource innovation-safe representation). *Let L be a finite source library containing s_0 , let the catalog and closure satisfy Assumption 3.2, and let $w_{s_0} = 0$ with exact rational $w_s > 0$ for every active source strategy. Then $\mathcal{P}_{\text{safe}}(L)$ is finite and nonempty, $\mathcal{M}_{\text{safe}}(L) \neq \emptyset$, and every $L' \in \mathcal{P}_{\text{safe}}(L)$ has the same finite-horizon productive values as L . Under the contraction clause of Assumption 3.2, it also has the same stationary productive value and complete optimal stationary-action correspondence.*

Proof. The source is feasible relative to itself and has only finitely many sublibraries, so an exact burden minimum is attained. Feasibility is exactly $K_{L'} = K_L$; the value and action conclusions follow from Theorem 3.3. Neither optimizer uniqueness nor a particular deletion path follows.

For comparison with observable process behavior, write $L \sim_{\text{DI}} L'$ when the libraries have equal frontier and equal availability-tagged project cost, duration, incumbent-reward, and joint terminal observations at every belief. Raw closure is *detectable* when equal frontiers but unequal closures necessarily change at least one such observation after generation, verification, admission, and projection.

Theorem 4.2 (Frontier-closure characterization). *Under Assumption 3.2, $F_L = F_{L'}$ and $C_L = C_{L'}$ imply $L \sim_{\text{DI}} L'$. If raw closure is additionally detectable in the preceding sense, the implication is an equivalence. The forward direction, which supplies productive preservation for every safe-feasible library, does not require detectability.*

Proof sketch. Equal frontier-closure states make every library-invariant primitive and its projected transition equal. Conversely, detectability rules out an unequal closure once all availability-tagged observations agree. Complete quotient and assumption-boundary arguments are in Online Resource 1, Section S2.

4.2 Identity closure induces a tagged cover

The reduction in this subsection is the computational consequence of the preservation semantics. Once the tagged universe is constructed, the resulting optimization problem is ordinary weighted set cover: its generic binary model, NP-hardness antecedent, greedy approximation theorem, mask recurrence, and standard cover preprocessing are not claimed as new combinatorial-optimization theory. The application-specific content is the exact mapping from frontier and closure equality to tagged coverage, together with its identity-closure limit.

IDENTITY-CLOSURE ALGORITHMIC BOUNDARY. The tagged rows, binary covering model, preprocessing rules, MIP, and all algorithms in Section 5 require $\text{cl}(A) = A$. The general semantic problem remains $F_{L'} = F_L$ and $\text{cl}(\bigcup_{s \in L'} \text{mods}(s)) = C_L$; it does not become an independent module-row cover when closure contains complementarities.

Assume throughout this subsection that closure is identity. Define the disjoint tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}, \quad (4.3)$$

and, for each $s \in L$, its source-relative coverage

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (4.4)$$

Frontier and module tags remain distinct even when their underlying identifiers print alike. Exact equality in the first term means that every tied source attainer covers the belief row.

Theorem 4.3 (Tagged-cover equivalence under identity closure). *Let B, M, S be finite, let $B \neq \emptyset$, and let $s_0 \in L' \subseteq L \subseteq S$. Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$; $j_{s_0} = 0$, $\text{mods}(s_0) = \emptyset$, maxima are attained, and closure is identity on 2^M . Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (4.5)$$

No unique frontier attainer or unique module carrier is assumed. If $F_L(b) = 0$, mandatory s_0 itself covers $\text{frontier}(b)$.

Proof. If L' preserves the frontier, a finite maximizer in L' covers each frontier tag. Under identity closure, $C_{L'} = C_L$ is equality of raw module unions, so some retained carrier covers every module tag. Conversely, source inclusion gives $F_{L'} \leq F_L$ and $C_{L'} \subseteq C_L$. Coverage supplies a retained source attainer at every belief and a retained raw carrier for every source module, giving the reverse inclusions. Ties and multiple carriers are existential choices, while the zero-frontier row is witnessed by s_0 .

For exact rational weights, the equivalent binary model is

$$\min_x \sum_{s \in L} w_s x_s, \quad (4.6)$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b) = F_L(b)} x_s \geq 1, \quad b \in B, \quad (4.7)$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (4.8)$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (4.9)$$

The source restriction is built into the columns. All zero-frontier rows may be removed after fixing s_0 , but retaining them in $\mathcal{U}_L^{\text{tag}}$ keeps the semantic construction literal.

The boundary is substantive. If $\text{cl}(\{a, b\}) = \{a, b, c\}$ although no strategy carries c , an independent row for each member of C_L rejects even the source library. General closure therefore requires direct closure equality or a separately proved extended formulation. Frontier rows alone and module rows alone are likewise insufficient; exact counterexamples are in Online Resource 1, Section S4.

4.3 Baseline mixed-integer formulation

Equations (4.6)–(4.9) are the baseline MIP. Its input is a validated, canonically ordered identity-closure instance with exact weights and declared mandatory columns. Exact preprocessing may fix columns and remove satisfied rows before JuMP and HiGHS receive the residual binary cover. Exact weights are scaled to an integer objective only after the scaling identity and solver range are verified.

The output is one incumbent in canonical original-strategy order together with the solver version, controls, termination/primal/dual statuses, objective, bound, reported gap, time, nodes when available, preprocessing audit, and complete log. The method makes no all-ties claim: a tied incumbent is one declared representative, and stable input ordering does not turn the solver’s choice into a theorem about optimizer identity. A returned vector is accepted only when every component is exactly 0 or 1; no fractional value is rounded. Reconstruction is followed by independent exact checks of mandatory retention, tagged coverage, the original frontier, the original identity closure, and burden. A HiGHS status of **OPTIMAL** remains solver evidence, not exhaustive or formal proof. When preprocessing leaves no residual row or column, the solver is recorded as not called rather than given a fabricated status.

4.4 Computational complexity

IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID) has the following explicit decision encoding. The input lists a finite source $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set, exact binary-rational profiles, a finite module universe and incidence rows, identity closure, positive binary-integer active weights, and a nonnegative binary-integer threshold D . Strategy s_0 is mandatory, has zero profile, no modules, and zero

weight. The question is whether some $s_0 \in L' \subseteq L$ preserves the exact source frontier and closure and has $W(L') \leq D$.

Theorem 4.4 (Identity-closure safe-compression complexity). *Under the explicit encoding in the preceding paragraph, ISC-ID is NP-complete. It remains NP-complete on the restricted subclass with one belief, zero profiles for every strategy, a zero-weight module-free inactive strategy, identity closure, a source whose active module rows have full union, and unit active weights. Consequently, minimum-burden identity-closure safe compression is NP-hard, and any broader model class containing this subclass is NP-hard. No NP-membership conclusion for arbitrary nonidentity closure follows unless its encoding admits a polynomial exact closure-equality verifier.*

Proof sketch. A selected-strategy bit vector is a polynomial certificate: exact rational maxima, row unions, binary-integer burden, source inclusion, and mandatory retention are polynomially checkable in the explicit input length. For hardness, normalize Karp’s SET COVER decision problem (Karp 1972) to a nonempty universe and a full-union indexed family, using fixed yes/no outputs for the empty- or uncovered-universe branches. Give each universe element a module, each indexed set an active strategy carrying its modules, every active strategy unit weight and zero profile, and add one zero-profile inactive strategy. Use one belief and copy the budget. The inactive strategy preserves the trivial frontier, while identity-closure equality is exactly set-cover equality and burden is exactly selected-set cardinality. Incidence size and integer bit lengths are copied polynomially. The complete membership, normalization, forward, reverse, size, and threshold audit is in Online Resource 1, Section S4. No strong-hardness, inapproximability, or APX-hardness claim is made.

4.5 Exact preprocessing of the tagged cover

Fix the tagged universe from the source before reducing the model. After declared mandatory strategies are selected, let Q be the unsatisfied rows, A the available nonfixed columns, and $\mathcal{R}_s^Q = \mathcal{R}_s \cap Q$. Mandatory weights are nonnegative, optional weights are strictly positive exact rationals, and there are no identity-sensitive side constraints, groups, exact-cardinality conditions, or mutual exclusions beyond the pure cover. All incidence and weight comparisons are exact.

Proposition 4.5 (Exact preprocessing suite). *Under the assumptions in the preceding paragraph, the following fixed-point rules preserve feasibility, the exact optimum value including forced-weight offsets, and at least one liftable optimum:*

1. fix the unique available carrier of any residual requirement;
2. delete nonmandatory s when $\mathcal{R}_s^Q \subset \mathcal{R}_t^Q$ and $w_t \leq w_s$;
3. for identical residual coverage, retain a stable minimum-weight representative and record every equal-weight substitution;
4. after a forced selection, remove satisfied rows and recompute all rules;
5. delete a nonmandatory column with empty residual coverage; and
6. merge rows having identical carrier sets.

Strictly cheaper dominance, strictly heavier duplicates, empty positive-weight columns, forced columns, and duplicate rows preserve all original optimizer identities. Equal-weight duplicate columns recover all identities through the recorded substitution map. Equal-weight strict dominance need not preserve all optimizer identities and makes no such claim. Restarting the stable scan after each transformation terminates after finitely many row or column removals and yields a fixed point. A zero-carrier residual row is an infeasibility certificate, not an invitation to repair the instance.

Proof sketch. A unique carrier is forced by its covering row. A no-heavier coverage superset can replace a dominated selected column without losing a witness. Identical columns permit exact substitution; identical rows impose the same inequality; and an empty positive-weight column occurs in no optimum. Forced columns contribute a selection-independent offset, and propagation composes these correspondences. Every successful step removes a finite row or column, which proves termination. The strict/equal weight cases give precisely the optimizer-identity qualifications in the statement. Full propositions, adversarial examples, reconstruction maps, and the deterministic rule order are in Online Resource 1, Section S5.

5 Algorithms and guarantees

All methods in this section consume the same validated identity-closure instance. Strategies and tagged requirements have canonical original order; weights are exact rationals; mandatory flags, source profiles, raw module memberships, the source frontier, and the source closure are stored independently of the incidence matrix. Preprocessing supplies an exact objective offset and a reconstruction map. The number of residual optional strategies is n , the number of residual tagged requirements is r , and strategy i has residual mask R_i and positive weight w_i .

APPLICABILITY BOUNDARY. Enumeration, requirement-mask dynamic programming, the tagged-cover MIP, weighted greedy, and the journal preprocessing pipeline solve only the identity-closure subclass established in Section 4.2. They are not algorithms for arbitrary closure. Rechecked deletion is semantically valid for a general declared closure only when every trial deletion is evaluated by a separate exact general-closure checker; the common implementation and benchmark use identity closure.

5.1 Common output and certificate

Every algorithm returns original strategy identifiers in canonical order, the exact burden, its declared completion status, its tie rule, an algorithm trace or search counters, and any required random seed. A result is reportable only after an independent certificate recomputes from the original catalog:

1. every mandatory strategy is retained;
2. every tagged requirement is covered;
3. the complete source operating frontier is preserved exactly;

4. the complete source identity closure is preserved exactly; and
5. the exact burden agrees with the returned burden.

Passing this certificate establishes feasibility and burden for the returned library. It does not establish heuristic optimality, convert a solver status into a proof, or show that two implementations are independent. Complete-tie output is permitted only when the preprocessing map declares every optimizer identity reconstructable; otherwise the result is one explicitly declared representative.

Table 1: Algorithm guarantees and claim boundaries. Guarantees originate in human proofs or transferred classical theorems, not benchmark timings.

algorithm	guarantee	evidence class	scope and limit
Enumeration	Exact optimum after complete finite search	Human proof + exact finite computation	Small finite identity-closure instances; exponential in optional strategies
Mask DP	Exact optimum; $O(n 2^r)$ recurrence operations	Human proof + exact finite computation	FPT in residual requirements r , not polynomial in full input size
HiGHS MIP	Exactly rechecked feasible incumbent; solver-optimal status reported separately	Mixed-integer solver evidence + exact finite post-check	OPTIMAL is neither formal proof nor exhaustive enumeration
Weighted greedy	$H(d) \leq H(r)$ weighted-cover approximation	Transferred primary-source theorem	Positive active weights and identity-closure tagged-cover equivalence only
Weighted greedy + reverse deletion	$H(d)$ guarantee inherited; deletion cannot increase burden	Transferred theorem + exact deletion check	Same identity-closure scope; reverse deletion adds no general-closure guarantee
Cardinality greedy	No burden approximation guarantee for heterogeneous weights	Algorithmic limitation	Unweighted score is reported as a comparator
Heaviest safe first	Exact feasibility and irreducibility only; registered family proves an unbounded burden ratio	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Lightest safe first	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Maximum immediate release	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Minimum unique-carrier exposure	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Declared source order	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked

algorithm	guarantee	evidence class	scope and limit
Random-order deletion	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
32-start random deletion	Best of 32 certified irreducible endpoints; no approximation factor	Synthetic algorithmic evidence + exact finite certification	Registered seeds and fixed 32 starts; not adaptive

5.2 Independent exact methods

5.2.1 Complete enumeration

Pseudocode: complete enumeration.

1. **Input:** a validated residual cover, forced selections and exact offset W_0 , a tie mode, and a finite output cap.
2. Visit every residual binary selection in increasing canonical mask order; test full tagged coverage and sum its weight exactly.
3. Keep the lower burden. On equality, retain every selection in complete mode or the minimum selected-index list in representative mode.
4. Lift forced selections and reconstruction classes, run the independent certificate, and return the optimizer set or declared representative, exact burden, number of candidates, tie status, and certificates.

Proposition 5.1 (Correctness of complete enumeration). *Let the input be a valid finite identity-closure tagged instance with exact nonnegative mandatory weights, strictly positive optional weights, at least one carrier for every residual requirement, and a valid preprocessing reconstruction map. Complete enumeration returns a minimum-burden safe source sublibrary after checking all 2^n residual selections. If complete ties are requested, the map preserves all optimizer identities, and the output cap is not exceeded, it returns every original optimizer; otherwise it returns the declared stable representative.*

Proof. Every admissible library is a forced set plus one residual subset. Enumeration evaluates every such subset and takes the exact minimum; the fixed offset and valid lift preserve burden and feasibility. The final certificate maps tagged feasibility back to the original source semantics through Theorem 4.3. Complete reconstruction follows only under the stated map and output-cap conditions. Time is $2^n \text{poly}(|I|)$, so enumeration is exact but exponential in the number of optional strategies.

5.2.2 Requirement-mask dynamic programming

Encode a residual requirement subset by an r -bit mask. With $D_i(q)$ the least exact burden that attains mask q using the first i strategies, the recurrence is

$$D_i(q) = \min \left\{ D_{i-1}(q), \min_{p: \text{ORR}_i = q} (D_{i-1}(p) + w_i) \right\}. \quad (5.1)$$

Pseudocode: requirement-mask DP.

1. **Input:** the same validated residual cover, exact offset, tie mode, and output cap used by enumeration.
2. Initialize $D_0(0) = 0$ and every other mask as unreachable.
3. For strategies in canonical order and masks in increasing order, emit the skip transition before the take transition $q \mapsto q \text{ OR } R_i$. Keep the lower exact burden.
4. On an exact tie, store the first predecessor under this traversal or every predecessor in complete-tie mode.
5. Read the all-ones mask, reconstruct, add W_0 , lift to original identifiers, certify, and return burden, selections, visited states, transitions, tie status, and certificates.

Theorem 5.2 (Exact requirement-mask dynamic programming). *Let the input satisfy every assumption of Proposition 5.1. The recurrence in (5.1) reconstructs at least one minimum-burden safe source sublibrary. Representative mode uses $O(n2^r)$ recurrence operations, $O(2^r)$ exact cost states, and $O(n2^r)$ predecessor records when full reconstruction is retained. Value-only two-layer storage uses $O(2^r)$ memory. Materializing T tied optimizers adds $O(Tn)$ output bits and may be exponential in n . Its bit time is $2^r \text{ poly}(|I|)$: the method is fixed-parameter tractable in r , but it is not polynomial in unrestricted input size.*

Proof sketch. Induct on i . At $i = 0$, only the empty subset attains mask zero. Every subset of the first i strategies either skips i , or removes i to give a predecessor mask and then takes it. The two transitions are exhaustive and construct valid subsets, so the recurrence stores the least exact burden for each mask. The all-ones mask is precisely residual feasibility. Offset, lifting, and Theorem 4.3 yield a safe global optimum. The state, transition, memory, and output counts give the displayed bounds. The complete induction and reconstruction proof is in Online Resource 1, Section S6.

5.3 Weighted greedy construction

After selecting every mandatory or forced strategy, let U^0 be the remaining tagged requirements and let

$$d := \max_i |R_i|, \quad H(d) := \sum_{j=1}^d \frac{1}{j}.$$

Pseudocode: weighted greedy and reverse deletion.

1. **Input:** a valid residual identity-closure cover and a Boolean reverse-deletion flag.
2. While requirements remain, choose a strategy minimizing $w_i/|R_i \cap U|$, comparing exact rationals; break ties by the lowest canonical original strategy index. Add it and delete its newly covered rows. If no carrier exists, return an explicit infeasibility failure.
3. If requested, examine the greedily selected optional strategies in reverse selection order and delete one only when complete tagged coverage and the original semantic checks remain true.

4. **Output:** the selected original library, exact burden, step scores, uncovered counts, reverse-deletion trace, tie declaration, and final independent certificate.

Theorem 5.3 (Transferred weighted-greedy guarantee). *Let the source be a finite identity-closure instance satisfying Theorem 4.3; let every residual requirement have a carrier; let mandatory weights be nonnegative and every optional weight be positive; and apply only mandatory/forced preprocessing with a valid exact offset and lift. If $U^0 \neq \emptyset$, weighted greedy returns a safe library of burden W_G satisfying*

$$W_G \leq H(d) \text{OPT} \leq H(r) \text{OPT}.$$

Reverse deletion returns a safe, inclusion-wise irreducible library with $W_{GR} \leq W_G$, so it satisfies the same bound. If $U^0 = \emptyset$, the forced library is optimal and the factor is one. The guarantee does not extend to arbitrary nonidentity closure.

Proof sketch. On the residual tagged universe, the rule is precisely the positive-cost weighted set-cover greedy algorithm of Chvátal (1979), which gives $G \leq H(d) \text{OPT}_R$. Adding a nonnegative fixed burden W_0 preserves the displayed factor. Reverse deletion accepts only feasibility-preserving removals of positive-weight strategies and therefore cannot increase burden. Later removals cannot restore coverage lost by an earlier rejected removal, so one reverse pass is irreducible. The theorem transfers the classical guarantee through Theorem 4.3; it does not reprove the classical set-cover theorem. The cardinality variant replaces w_i by one in the score and has no retention-burden approximation guarantee when weights differ.

With straightforward incidence scans, greedy and reverse deletion each use a polynomial number of exact operations (at most n selection/deletion rounds and $O(nr)$ incidence work per round). This implementation bound does not alter the approximation factor.

5.4 Certified rechecked deletion

Rechecked deletion starts from the full source and never infers safety from a stale certificate. At every scan, each nonmandatory trial deletion is checked against the original source frontier and closure; after an accepted deletion, all candidate certificates are discarded and the scan restarts.

Pseudocode: rechecked deletion.

1. **Input:** a validated source library, one declared deterministic score or a seeded stable random order, and an exact feasibility checker.
2. Recompute the safety of every currently retained nonmandatory one-strategy deletion. If none is safe, stop.
3. Choose among safe candidates by the declared score and tie rule, append the exact check to the trace, accept the deletion, and restart.
4. Perform a complete final scan. Return the endpoint, exact burden, trace, frontier/closure check counts, complete-scan count, seed and order when stochastic, feasibility certificate, and irreducibility certificate.

The deterministic scores are:

- heaviest-safe-first, with exact weight ties broken by canonical index;
- lightest-safe-first, with the same tie rule;
- maximum immediate burden release, which equals heaviest-safe-first under additive burden;
- minimum remaining unique-carrier exposure, defined for safe deletion s from retained set A by

$$E(s \mid A) := \left| \left\{ u : \sum_{t \in A \setminus \{s\}} \mathbf{1}\{u \in \mathcal{R}_t\} = 1 \right\} \right|,$$

with ties resolved by greater exact burden release and then canonical index; and

- declared source order, using a complete caller-supplied permutation or canonical order by default.

The exposure score uses only source incidence and no held-out financial quantity. Random-order deletion uses the first safe strategy in a recorded stable seeded permutation.

Maximum immediate release is retained as an API alias and as a historical label in the locked benchmark because those raw records are immutable. It is not an independent scientific baseline, and coincident results under the two names are not corroborating replications.

Proposition 5.4 (Certified deletion endpoint). *Let the source be finite and safe relative to itself, let all mandatory strategies be protected, and suppose every accepted trial deletion is recomputed exactly and shown to preserve the original frontier, declared closure, and burden identity. Then every deterministic or seeded-random rechecked-deletion variant above terminates at a safe source sublibrary. A complete exact final scan certifies that no retained nonmandatory strategy can be safely deleted. This is a one-deletion irreducibility certificate, not a global-optimality or approximation certificate.*

Proof. The source is safe. Exact certification of each accepted trial preserves safety inductively, and each step removes one strategy, so the trace is finite. The final scan checks every remaining eligible deletion. The score and random order choose only among already certified candidates and therefore do not change the argument. If closure is nonidentity, this proposition requires the separate exact closure checker stipulated in its assumptions.

5.4.1 Multi-start random deletion

Pseudocode: multi-start deletion. Run rechecked random-order deletion for a declared number p of independently derived, recorded seeds. Retain the certified endpoint of lowest exact burden; break endpoint ties by the lexicographically smallest selected-index tuple and then the earliest start. Return every seed and endpoint summary, the winning trace, aggregate check counts, and the winner’s feasibility and irreducibility certificates. The method costs p certified deletion runs and provides no

approximation factor; additional starts cannot worsen the best observed burden, but they do not prove optimality.

5.5 Worst-case local–global separation

Theorem 5.5 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Use one belief, identity closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy with zero profile, zero burden, and no modules. For each i , let singleton s_i have zero profile, weight one, and module set $\{m_i\}$. Let bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. Rechecked heaviest-safe-first deletes the uniquely heaviest safe bundle, terminates at the k -singleton library of burden k , and returns an inclusion-wise irreducible endpoint. The unique global optimum retains only the bundle and has burden $1 + \epsilon$. The exact ratio is*

$$\frac{k}{1 + \epsilon},$$

so heaviest-safe-first and one-deletion irreducibility alone have no constant burden guarantee over this family. The conclusion does not apply to every deletion order or every certified deletion heuristic.

Proof. All profiles are zero, so safety is exactly coverage of all modules. At the source, the bundle makes every singleton deletable and all singletons together make the bundle deletable. Since $\epsilon > 0$, the bundle is uniquely heaviest and is deleted first; afterward each singleton uniquely carries its module. Any safe library without the bundle must therefore retain all singletons and costs k , whereas the bundle alone is safe and costs $1 + \epsilon < k$; positive weights make it the unique optimum. Fixing, for example, $\epsilon = 1/2$ makes the ratio unbounded as k grows. A singleton-first order instead reaches the bundle optimum, proving the stated order boundary. Full finite fixtures and the reduction to the earlier small example are in Online Resource 1, Section S7.

5.6 Guarantee and evidence boundary

Enumeration and mask DP are exact finite methods under their stated identity-closure assumptions; the former is exponential in n , and the latter is exponential in r . Weighted greedy is approximate but has the transferred $H(d)$ bound; reverse deletion preserves that bound. Rechecked deletion and multi-start return exactly certified feasible, irreducible endpoints but remain heuristics. The MIP in Section 4.3 supplies solver evidence plus an exact returned-library postcheck. Complete proofs and machine-readable certificate schemas are in Online Resource 1; benchmark runtimes and method comparisons are experimental evidence reported separately in Section 7.

The suite does not include a certified multi-exchange neighborhood or an LP relaxation baseline. Accordingly, the local–global theorem isolates the named heaviest-safe-first rule and one-deletion irreducibility; it says nothing about stronger local search. Adding those baselines after seeing the locked benchmark would be post hoc and is reserved for a separately registered study.

6 Economic consequences of unsafe and capacity-constrained retention

The preservation constraint has an economic purpose beyond reproducing a static payoff menu. Frontier equality protects current operating opportunities; closure equality protects the inputs from which future opportunities can be generated. This section first separates those two value channels and then studies two outer resource decisions: retention under a hard capacity and retention under a unit burden price. Neither outer problem changes the generation, verification, admission, or positive-duration timing inside the productive process of Section 3.

6.1 Operational and generative value

For a finite horizon, let $P_n(b, L)$ be the value when the library is frozen and research is unavailable, and let $U_n(b, L)$ be the corresponding full value when every feasible positive-duration project is available. Thus

$$\begin{aligned} P_0(b, L) &= 0, \\ P_{n+1}(b, L) &= F_L(b) + \beta \sum_{b'} P_B(b, b') P_n(b', L), \end{aligned} \quad (6.1)$$

where P_B is the filtered-belief transition under continued operation. Write $\Omega_n = U_n - P_n$ for the research-option premium. For inserting a candidate s , define total, operational, and generative increments by

$$\begin{aligned} \mathcal{I}_n(s \mid b, L) &:= U_n(b, L \cup \{s\}) - U_n(b, L), \\ \Delta_n^{\text{op}}(s \mid b, L) &:= P_n(b, L \cup \{s\}) - P_n(b, L), \\ \Delta_n^{\text{gen}}(s \mid b, L) &:= \Omega_n(b, L \cup \{s\}) - \Omega_n(b, L). \end{aligned} \quad (6.2)$$

Proposition 6.1 (Operational–generative decomposition). *For every finite horizon n , filtered belief b , finite raw library L , and admissible candidate s ,*

$$\mathcal{I}_n(s \mid b, L) = \Delta_n^{\text{op}}(s \mid b, L) + \Delta_n^{\text{gen}}(s \mid b, L). \quad (6.3)$$

If $F_{L \cup \{s\}} = F_L$, then the operational increment is zero. If in addition $C_{L \cup \{s\}} = C_L$, then the total increment is zero under the projection assumptions of Theorem 3.3.

The equality is an accounting identity, not a sign theorem. A strategy may be silent in the passive operating problem and valuable because it carries a module that enables research; conversely, the generative residual need not be positive if insertion changes costs or the relative appeal of projects. The frontier–closure projection makes the two channels operationally useful: frontier silence identifies zero current operating value, whereas joint frontier and closure silence identifies zero productive value within the declared process.

6.2 Sharp loss from frontier-only retention

Consider a dominated bridge strategy that uniquely carries the capability needed for one descendant project. Its deletion leaves the source frontier unchanged but makes the distinguished descendant unreachable. The following statement fixes the scope of the loss calculation: the incumbent operating block is common, there are no other differential projects or continuation gains, and the terminal continuation gain is bounded by C .

Theorem 6.2 (Sharp normalized frontier-only bridge loss). *Fix exact rationals $0 \leq \beta < 1$, $0 \leq \rho, \pi \leq 1$, a reward cap $C \geq 0$, a project cost $\kappa \geq 0$, and an integer delay $d \geq 1$. Suppose the bridge uniquely enables the descendant, retention gives survival mass ρ^d and conditional admission probability π , deletion makes that descendant mass zero, and $\kappa \leq \beta^d \rho^d \pi C$. Then the retained-minus-deleted value loss is*

$$\text{Loss}_d = \beta^d \rho^d \pi C - \kappa. \quad (6.4)$$

For fixed $(\beta, \rho, \pi, d, \kappa)$, this is the sharp upper bound over descendant rewards in $[0, C]$, attained at C . Whenever the denominator is positive, the loss divided by the attainable net bridge value equals one. If $C \leq 1$, the additive loss is at most one; arbitrarily large additive loss requires reward scaling.

The proof expands the one available project branch. Retention contributes the discounted joint survival-and-admission mass times the capped continuation gain, less cost; deletion contributes zero. Maximization over the reward interval is attained at its upper endpoint. The factor $\rho^d \pi$ comes from sequential generation and conditional admission, not from an independence assumption. Common operating value cancels under frontier equality; if operation differs while research is in progress, its exact discounted difference must be added. Online Resource 1 gives the full fixture, proof, and delay qualifications.

The result explains why operational redundancy is not generative redundancy. A frontier-only rule can be exact for today's operating menu and still remove the entire positive net value of a future bridge. It does not assert an unbounded normalized loss or a general value formula for arbitrary deleted libraries.

6.3 Calibrated closure-option mechanism experiment

The theorem is universal under its assumptions, but it does not say whether a finite-data policy can recognize a valuable bridge. We therefore isolate that question in a design-locked synthetic experiment. The innovation-safe and frontier-only libraries have identical current frontiers and identical retention burdens. Only the safe library carries the capability needed to run one declared research project. For delay (d), survival probability ρ , admission probability π , descendant gain G , and research cost κ , the registered treatment margin is

$$M = \beta^d \rho^d \pi G - \kappa.$$

Regime	True margin	Oracle mean [95% CI]	Learned mean [95% CI]	Adoption
Adverse margin	-0.0270	0.0000 [0.0000, 0.0000]	-0.0000 [-0.0000, 0.0000]	0.05%
Null margin	0.0000	0.0000 [0.0000, 0.0000]	-0.0002 [-0.0003, 0.0000]	5.76%
Low positive dose	0.0025	0.0026 [0.0019, 0.0032]	0.0001 [-0.0001, 0.0002]	7.35%
Powered positive dose	0.0540	0.0541 [0.0529, 0.0554]	0.0512 [0.0500, 0.0525]	94.17%

Table 2 Calibrated closure-option mechanism experiment, with 4096 synthetic worlds per prespecified regime. Current frontier and burden are equal by construction; only the innovation-safe library can admit the designated bridge descendant. Intervals are registered finite-simulation uncertainty summaries. This is synthetic mechanism evidence, not a market-return, alpha, forecasting, or causal estimate.

The safe oracle researches exactly when $M > 0$; the frontier-only project is infeasible. A learner observes an independent noisy proposal signal and adopts only when it exceeds a threshold calibrated under the null.

The noise law uses only aggregate nuisance estimates from 2479 complete proposal paths and 623647 within-cell residual observations in the point-in-time financial study. Cross-strategy date means are removed before estimating volatility, lag-one dependence, and tail thickness. These quantities calibrate noise and dependence, not the sign of any regime. The test-world seeds are disjoint from calibration, and the powered margin is fixed at twice the null threshold before test simulation. Thus its size is a power-design dose, not an estimated return premium.

Table 2 reports all four prespecified regimes. At the low positive dose, the oracle value is positive but the learned-value interval crosses zero and adoption is only 7.35%. At the powered dose $M = 0.0540$, the learned mean is 0.0512 and adoption is 94.17%. The null point adoption rate, 5.76%, satisfies the registered 6.5% finite-test gate; its Wilson upper endpoint is 6.52%, so the interval is not wholly below that threshold. Every world passes the equal-frontier, equal-burden, and bridge-availability structural checks.

This evidence makes the option/exercise distinction concrete. Exact closure preservation can create access to a genuinely positive project while a conservative learner rationally fails to exercise a small option at the available signal-to-noise ratio. Conversely, the powered row shows recovery only for a deliberately larger treatment. The experiment validates the declared bridge mechanism under calibrated synthetic noise. It is not a historical market test, a causal return estimate, an alpha estimate, or a claim that the powered margin is empirically prevalent.

6.4 Capacity and price extensions

Exact source-relative compression holds productive state fixed. A different managerial problem allows productive value to vary under a hard maintenance capacity or a unit burden price. For a finite eligible family, the hard-capacity value is a nondecreasing step function of capacity; complementarity can make it nonconcave and can leave a capacity-optimal library unsupported by any scalar price. The price-penalized value is the upper envelope of finitely many affine branches, hence is continuous, nonincreasing, convex, and piecewise affine, with optimizer burden weakly decreasing in price.

These facts are standard finite resource-allocation geometry rather than a second principal contribution. Complete assumptions, theorem statements, proofs, and counterexamples are moved to Online Resource 1, Section S9. The main article uses only their implication: local safety, global minimum burden, and value under an insufficient capacity are three different questions.

7 Registered computational study

Registered Algorithmic Compression Benchmark v2 is a finite synthetic study of tagged identity-closure instances. It can test exact-method agreement, certified burden quality, preprocessing prevalence, and the declared adversarial mechanisms. It cannot establish average-case complexity, steady-state runtime rankings, or a population law for financial libraries.

7.1 Registered design, execution, and audit

The final registry contains 173 analysis instances: 24 small exact, 128 structured, and 21 adversarial. Family B requested variation in strategy and tagged-row counts, coverage density, module overlap, bundle prevalence, unique-carrier frequency, weight dispersion, and frontier-module association. Family C contains the predeclared gap, tie, duplicate, dominance, rare-carrier, symmetry, and bundle mechanisms. Failed generation remains a terminal result; no failed row is silently repaired or dropped.

The algorithm grid contains enumeration, requirement-mask DP, two registered HiGHS formulations, weighted and cardinality greedy, greedy plus reverse deletion, rechecked deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate additive burden release is exactly the heaviest-safe-first ordering. Both labels remain in the immutable registered records, but they are one method rather than independent scientific baselines.

The audit passed all 3907 registered run units and accepted 3526 returned libraries after original-coordinate mandatory, coverage, frontier, identity-closure, and exact-burden checks. It retained 381 unsuccessful records in their terminal categories. HiGHS OPTIMAL remains solver evidence plus an exact incumbent check, not formal or exhaustive proof.

7.2 Exact agreement and applicability

Among 38 registry instances, all required exact-method burdens were available and equal on 31; 7 did not have a complete comparison. 6 adversarial enumeration jobs are recorded as IMPLEMENTATION_ERROR, but their substantive cause is known: the residual optional count exceeded the declared enumeration cap. Their historical statuses are not rewritten. They are enumeration-inapplicable under that cap, and exact agreement is unavailable rather than inferred. Optimizer identities differed at equal burden on 9 complete comparisons, which records valid tied optima rather than disagreement.

These are exact finite facts about specified instances. The medium structured references are HiGHS-optimal incumbents that pass exact feasibility and burden rechecks; no second solver was pinned.

Algorithm	Attained/reference	Mean gap	Median	q90	Worst
<i>Family A: exact finite reference</i>					
Weighted greedy	16/23	0.08684	0	0.25	0.5
Cardinality greedy	13/23	0.6375	0	1.4	7
Weighted greedy + reverse deletion	20/23	0.02923	0	0.16	0.25
Heaviest safe first	22/23	0.008696	0	0	0.2
Lightest safe first	5/23	1.586	1	3.32	7
Maximum immediate release	22/23	0.008696	0	0	0.2
Minimum unique-carrier exposure	17/23	0.2303	0	0.2686	3
Declared source order	11/23	0.6965	0.2857	2	3
Random-order deletion	8/23	1.384	0.6667	3.3	7
32-start random deletion	23/23	0	0	0	0
<i>Family B: HiGHS-OPTIMAL, exactly rechecked reference</i>					
Weighted greedy	82/111	0.02982	0	0.06452	0.5
Cardinality greedy	24/111	0.8904	0.25	2	7.5
Weighted greedy + reverse deletion	103/111	0.01038	0	0	0.5
Heaviest safe first	94/111	0.09983	0	0.2	3
Lightest safe first	11/111	4.128	0.7143	15	63
Maximum immediate release	94/111	0.09983	0	0.2	3
Minimum unique-carrier exposure	98/111	0.04435	0	0.05556	1
Declared source order	36/111	1.533	0.1818	5	20
Random-order deletion	28/111	1.753	0.2222	6.8	17
32-start random deletion	76/111	0.1324	0	0.5	2.667

Table 3 Registered heuristic burden quality. Relative gap is $(W - W^*)/W^*$. Attainment denominators include every registered row with an available reference; gap moments use exactly feasible returned libraries. Family A and Family B have different reference evidence and are deliberately not pooled.

7.3 Certified burden quality

Table 3 separates Family A exact finite references from Family B exactly rechecked HiGHS-optimal references. Within those distinct evidence classes, weighted greedy plus reverse deletion and heaviest-safe-first often attain the available reference, while several other deletion orders have larger observed tails. Multi-start attains every available Family A reference but not every Family B reference. These are descriptive outcomes of the registered instances, not approximation guarantees or an algorithm ranking beyond the design.

The registered heaviest-safe-first gap family gives a different kind of evidence. Its exactly rechecked finite ratios match Theorem 5.5 at every registered scale. The theorem establishes the universal family statement; the experiment checks the implementation on the named fixtures.

7.4 Preprocessing and realized design limits

Full fixed-point preprocessing has a strong realized effect. In every structured size cell, median variable and requirement reduction is complete among available records. This is evidence that forced-carrier propagation is important for these generators. It is simultaneously a limitation: most structured instances do not exercise a difficult residual MIP or DP search, so the study cannot identify general residual-search scaling.

Scale	Parameter	Reference evidence	Observed ratio	Theoretical ratio
small	$k = 4, \epsilon = \frac{1}{2}$	exact finite	$\frac{8}{3}$	$\frac{8}{3}$
medium	$k = 16, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{32}{3}$	$\frac{32}{3}$
large	$k = 64, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{128}{3}$	$\frac{128}{3}$

Table 4 Registered adversarial instances for heaviest-safe-first. Equality between observed and theoretical ratios is a synthetic fixture check of the human-proved family, not an empirical approximation guarantee.

Size	Preprocess n	Median var. red.	Median row red.	Median log PAR-2 ratio	32-start complete	Improved
N64-R32	28/32	1	1	1.187	28/32	20/28
N64-R64	28/32	1	1	1.29	28/32	19/28
N192-R32	28/32	1	1	2.031	15/32	12/15
N192-R64	28/32	1	1	2.243	27/32	21/27

Table 5 Registered preprocessing and multi-start evidence on structured instances. Reductions are shares removed. The log ratio is $\log(\text{PAR2}_{\text{full}}/\text{PAR2}_{\text{mandatory}})$, so a positive value favors the mandatory-only pipeline in elapsed penalized runtime despite the model-size reduction. Unavailable multi-start endpoints are not coded as zero improvement.

The registered generator also contains infeasible requested cells. All 352 run units arising from 16 structured source-instance cells are retained, and the artifact audit shows that the failures remove complete combinations of low density and bundle/unique-carrier settings rather than occurring at random. Effects and interactions requiring those absent cells are not estimable from this design.

Multi-start deletion completes all registered starts on 143 instances and improves on the first start on 96. Memory-limited primary runs remain failures; successful sensitivity records are not substituted into their denominators.

7.5 Why runtime and registered regressions are not inferential evidence

The v2 worker timed first-call Julia execution without an algorithm-specific warm-up. Compilation can constitute nearly all recorded algorithm time in representative raw records. Execution also used 8 concurrent one-thread subprocesses on a dirty worktree, so cache, memory-bandwidth, launch wave, and compilation effects are part of each observation. The saved wall times remain auditable cold-start concurrent records, but the article does not use them to rank algorithms or infer steady-state scaling.

The registered performance profile also combines unlike success events: heuristics need an exactly feasible candidate, whereas exact methods must complete search. It is therefore excluded from the submission’s comparative evidence. A valid future study must separate feasibility construction, quality-qualified attainment, and exact completion.

The prespecified regressions used HC3 uncertainty on repeated algorithm and preprocessing rows from the same instance. HC3 addresses heteroskedasticity, not within-instance dependence. Complete generator cells are also absent. Accordingly, their

coefficients, intervals, and adjusted probabilities are preserved in Online Resource 1 as an executed-analysis record but are not used for inferential or structural-driver claims. No unregistered clustered model is substituted after seeing the results.

7.6 Failure handling and next experimental boundary

No final run ended at `TIME_LIMIT`, so v2 supplies no empirical distribution of timeout incumbent gaps. Generation failures, memory limits, interruptions, missing primals, and nonbinary-candidate rejections remain in the machine-readable census and all applicable denominators.

A stronger performance study requires a new registration rather than an amendment to v2: warm-up outside timing, repeated blocked measurements on a clean tree, prospective generator-feasibility checks, nonempty residual cover families, common success targets, instance-level paired or clustered analysis, prevalidated method applicability, and bounded-exchange and LP-bound baselines. No outcome from such a study is claimed here.

8 Financial evidence: point-in-time portfolio-library retention

The financial study asks whether innovation-safe compression can reduce the active validation burden of portfolio-strategy libraries while preserving a larger menu of admissible future research options than a budget-matched frontier-only library. It is a retrospective development-facing panel, not an independent confirmatory market test. Its contribution is the combination of point-in-time source construction, an explicit portfolio grammar, protected proposal decisions, exact structural rechecks, and a transparent held-out null result.

8.1 Sequential information boundary

At each annual origin y , early historical blocks construct the source portfolio-strategy library and its operating frontier; the subsequent predecision block estimates compression profiles and solves the safe and frontier-only arms; year $y + 1$ enumerates closure-enabled proposal options and freezes both policy choices; and year $y + 2$ supplies the held-out portfolio return paths. Proposal outcomes do not enter compression. Evaluation outcomes do not enter compression, option generation, uncertainty estimation, adoption, or any policy choice. The full candidate ledger retains unavailable and rejected candidates rather than conditioning the analysis on successful selection.

The primary universe is formed from the most liquid eligible US common equities at each compression origin. A separate plain-ETF universe is the registered replication when its point-in-time eligibility gate passes. Eligibility never requires survival into the proposal or evaluation window; observed delisting returns are retained and a terminated position subsequently moves to cash. The single failed ETF universe gate remains in the registered denominator. Raw security identifiers, prices, and returns remain inside the licensed boundary.

Universe	Origins	Mean source burden	Mean safe burden	Mean reduction	Added options	Held-out CE difference
Common equity	19	8169.2	2183.8	73.1%	65.95	0.0000
Plain ETF	18	8909.2	2231.7	74.7%	66.06	-0.0022

Table 6 Point-in-time portfolio-library retention results. Burden is the registered validation-computation index, not currency. Added options are safe-library proposal actions absent from the budget-matched frontier-only library. The held-out column is innovation-safe minus frontier-only certainty-equivalent difference: all 19 common-equity choices coincide, while one of 18 complete ETF choices differs. Safe solutions pass exact original-instance frontier, closure, inactive-retention, and burden rechecks; solver-reported optimality remains solver evidence.

Each source library applies a fixed factorial grammar of portfolio-level signals, filters, holding horizons, allocation rules, gross-exposure targets, and exits to the same contemporaneous universe. Closure consists of 31 declared rule, interface, data, corporate-action, portfolio-aggregation, and cost/capacity capabilities. These declarations are more operationally concrete than free-form module tags, but they remain a modeled grammar: they do not prove that tacit knowledge or every institutional artifact is frictionlessly recoverable.

8.2 Compression and protected proposal policy

The innovation-safe arm minimizes the registered validation-computation burden subject to exact source-frontier and full declared-closure equality. Its frontier-only comparator is augmented under the safe arm’s realized burden cap using a frozen preproposal order. The comparator action set is a subset of the safe action set in every gate-passed cell. Cash is always feasible, and the safe policy begins with the exact comparator choice. It can replace that choice only when a simultaneous moving-block lower confidence bound for incremental proposal value clears the registered economic hurdle. The policy therefore treats closure as an option, not a forced trade.

Across the gate-passed origins, 3 safe solutions are resolved by exact preprocessing and 34 are returned with HiGHS OPTIMAL status. Every result is independently reconstructed and rechecked for mandatory inactive retention, exact table frontier equality, exact identity-closure equality, and exact integer burden. Those rechecks certify feasibility and burden; for the solver-resolved cells, global optimality remains mixed-integer solver evidence rather than a formal or exhaustive proof.

Table 6 gives the aggregate structural and held-out results. Innovation-safe retention reduces the mean source burden index by 73.1% in common equities and 74.7% in ETFs while preserving all registered capabilities. Relative to the budget-matched frontier-only library, it exposes on average 65.95 additional common-equity proposal actions and 66.06 additional ETF actions.

8.3 Held-out result: options are not exercises

The protected policy makes 0 common-equity replacements. Consequently, every common-equity held-out innovation-safe-minus-frontier-only contrast is exactly zero because the two policies choose the same action. This is not evidence that the two

libraries are economically equivalent; it is evidence that none of the additional options clears the proposal-time uncertainty rule.

The ETF replication makes 1 replacement. That replacement realizes a held-out certainty-equivalent difference of -0.0390, producing a mean of -0.0022 over the complete ETF origins. All other complete ETF pairs choose the same action. The negative observation is retained and is not offset by reporting a favorable post hoc selector. Full origin-level results, the prespecified robustness grid, negative-control policies, search-bias diagnostics, and capacity records appear in Online Resource 1.

The contrast is a postdecision portfolio certainty-equivalent diagnostic under the registered risk and cost rule. It is not an alpha estimate, a forecast, or a deployable-performance claim. In particular, the many structural options do not count as independent financial observations, and the safe policy’s abstention cannot be reinterpreted as positive performance. The calibrated mechanism experiment in Section 6.3 supplies the separate finite-simulation result: a small positive bridge can be structurally real yet statistically unrecoverable, while a prospectively powered bridge is recovered.

8.4 Robustness, licensed-data boundary, and managerial implication

The proposal robustness analysis retains the full prespecified grid, including unavailable cells. Common-equity safe and comparator choices remain identical throughout that grid. ETF differences remain confined to the same origin and disappear at higher transaction-cost settings. A closure-sham analysis that was not frozen before proposal outcomes became known is explicitly unavailable, not reconstructed post hoc. Capacity analysis scores frozen choices without reselection and retains missing-source, incomplete-path, and hard participation-cap failures in its denominators.

Licensed daily panels were used locally to construct the point-in-time universes and portfolio paths. The public artifact redistributes only aggregate structural manifests, cryptographic seals, permitted summaries, and synthetic records. It contains no raw CRSP/WRDS rows, security identifiers, or return paths. Independent replication of the licensed stages therefore requires separate data access, whereas all published table values can be regenerated from the committed public-safe manifests.

For research governance, the result is useful precisely because it is not a success story about returns. A manager can remove substantial declared validation burden while exactly preserving current opportunities and a much larger research-option menu. Whether any added option should be exercised is a separate statistical decision, and a protected selector can rationally abstain. Removal from the active maintenance set also does not authorize deletion from legal, compliance, or reproducibility archives. A stronger empirical claim would require a newly locked market vintage or independent panel, documented real capability inventories, and monetary maintenance costs calibrated without using postdecision outcomes.

9 Conclusion

Financial strategy libraries can be treated as productive inventories rather than merely lists of currently attractive rules. Under the article’s explicit factorization assumption

for a declared finite partial-information process, the operating frontier and generative closure are a sufficient library coordinate. Their exact equality then provides a precise source-relative preservation contract. The result is conditional: if later transitions, admission, duration, or rewards depend on raw provenance or identities beyond those objects, the projection does not apply.

That semantic reduction leads to a distinct operations-research problem. Under identity closure, safe compression is a tagged weighted cover; even a restricted decision subclass is NP-complete. This complexity result does not make set cover novel, nor does it extend the tagged formulation to arbitrary closure. It explains why globally minimum retention burden cannot generally be inferred from local deletion safety. The parameterized gap family sharpens that distinction for a named heaviest-safe-first rule without claiming that every deletion order fails.

The algorithmic response is deliberately plural. Enumeration and requirement-mask dynamic programming provide independent exact finite methods on small instances; exact preprocessing can expose forced structure; the MIP workflow supplies auditable solver evidence; weighted greedy transfers a classical covering guarantee within the identity-closure boundary; and certified deletion methods provide feasible, irreducible endpoints without an exactness claim. Every route ends with an original-instance certificate, so feasibility does not depend on trusting a heuristic or solver status.

The registered synthetic benchmark supplies exact-method agreement, certificate, heuristic-burden, failure, and preprocessing evidence. Its cold Julia startup timings, unequal success targets, sparse structured grid, and within-instance dependence do not support algorithm-speed rankings or general scaling inference. These are locked finite-design findings under one concurrent, dirty-worktree execution environment, not average-case complexity or population evidence about financial libraries.

The retrospective point-in-time financial study illustrates the burden and option implications for an explicit portfolio grammar. Innovation-safe retention preserves every declared capability while materially reducing the validation-computation index and exposing many proposal actions unavailable to a budget-matched frontier-only library. Yet the protected policy never replaces the common-equity comparator, and its one ETF replacement loses held-out value. The calibrated synthetic experiment resolves the apparent tension: a small positive bridge may be structurally present but statistically unrecoverable, whereas a prospectively powered bridge is recovered under the declared noise law. The first result is retrospective financial evidence; the second is synthetic mechanism evidence. Neither supports a causal, forecasting, alpha, or deployable-performance claim, and licensed source rows remain outside the public artifact.

Two extensions are now most important. Algorithmically, nonidentity closure requires a representation that preserves derived capability interactions without replacing them by independent raw rows. Empirically, the next study should use a newly locked market vintage or an independent panel, richer documented institutional capability artifacts, and monetary maintenance costs calibrated independently of postdecision outcomes. Neither extension can be obtained by relabeling the present locked designs. Both should retain the separation among human proof, formal verification, exact computation, solver evidence, synthetic experiments, and retrospective financial evidence used here.

Supplementary information

Online Resource 1, entitled *Proofs, Extended Results, Computational Records, and Reproducibility Materials*, contains the complete supporting proofs, extended theory and counterexamples, canonical numerical records, the historical frozen $N = 1024$ study, the distinct registered algorithmic benchmark, point-in-time financial and calibrated mechanism records, solver and exactness records, theorem–evidence correspondence, and reproducibility commands. It compiles independently from `journal/aor/online_resource/main.tex`; raw licensed data are excluded. The complete SSRN upload PDF appends this resource after the article’s references; its pages and numbered results use an S prefix. Separate article and supplement PDFs are also provided in the repository’s `release/v0.2.0-ssrn/` directory.

Statements and Declarations

Funding. The author received no funding for this work.

Competing interests. The author declares no financial or non-financial competing interests.

Ethics approval and consent to participate. Not applicable. The study does not involve human participants or animals.

Consent for publication. Not applicable.

Data availability. Exact fixtures, synthetic-study inputs and public-safe aggregate outputs, generated tables, and generated figures supporting the journal version are provided in the project repository. The retrospective financial audits use licensed CRSP/WRDS source data. Raw and row-level licensed observations cannot be redistributed. Researchers with independent licensed access can reproduce the licensed-data stages using the documented input contracts. Public materials and data-access instructions are available at <https://github.com/dramirez37/financial-strategy-library-compression>. Original manuscript and documentation materials are licensed under CC BY 4.0, subject to the licensed-data and third-party exclusions in the repository’s LICENSE file.

Code availability. The project repository contains Julia 1.12.6 and Lean 4.32.0 source, pinned environments, exact fixtures, registered configurations, and reproducibility commands. The editable article and Online Resource source trees include every public-safe table, figure, and canonical-data input they consume; public-safe Julia and Lean source remains in the project repository linked above. Original software is licensed under the MIT License. The `release/v0.2.0-ssrn/` directory records PDF checksums, source provenance, and submission metadata; no archival DOI is claimed.

Author contributions. David Ramirez conceived the study, developed the theory and software, designed and conducted the analyses, curated the public artifacts, and wrote, reviewed, and approved the manuscript.

Generative-AI disclosure. OpenAI ChatGPT and Codex were used during manuscript preparation to assist with LaTeX drafting and editing, software development and review, literature-metadata checking, and repository-based analysis and reproducibility tasks. The author reviewed, revised, and verified all AI-assisted material and remains solely responsible for the manuscript, software, analyses, citations, and conclusions. No AI system is listed as an author, and no generative-AI image is used in the article or Online Resource.

Preprint disclosure. An earlier preprint candidate is preserved in the project history as release `v0.1.1-arxiv`; the identifier records an intended submission route, not an arXiv posting. The present revised and expanded version is prepared for SSRN as `v0.2.0-ssrn`. It has not been peer reviewed. No SSRN identifier or DOI is claimed before assignment.

References

- Allen, F., Karjalainen, R.: Using genetic algorithms to find technical trading rules. *Journal of Financial Economics* **51**(2), 245–271 (1999) [https://doi.org/10.1016/S0304-405X\(98\)00052-X](https://doi.org/10.1016/S0304-405X(98)00052-X)
- Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency: Supervisory Guidance on Model Risk Management. Supervision and Regulation Letter SR 11-7, attachment. Official supervisory guidance (2011). <https://www.federalreserve.gov/boarddocs/srletters/2011/sr1107a1.pdf>
- Bowers, M., Olausson, T.X., Wong, L., Grand, G., Tenenbaum, J.B., Ellis, K., Solar-Lezama, A.: Top-down synthesis for library learning. *Proceedings of the ACM on Programming Languages* **7**(POPL), 1–32 (2023) <https://doi.org/10.1145/3571234>
- Baker, N.R., Souder, W.E., Shumway, C.R., Maher, P.M., Rubenstein, A.H.: A budget allocation model for large hierarchical R&D organizations. *Management Science* **23**(1), 59–70 (1976) <https://doi.org/10.1287/mnsc.23.1.59>
- Chvátal, V.: A greedy heuristic for the set-covering problem. *Mathematics of Operations Research* **4**(3), 233–235 (1979) <https://doi.org/10.1287/moor.4.3.233>
- Cassandra, A.R., Littman, M.L., Zhang, N.L.: Incremental pruning: A simple, fast, exact method for partially observable markov decision processes. In: *Proceedings of the Thirteenth Conference on Uncertainty in Artificial Intelligence*, pp. 54–61. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA (1997). <https://arxiv.org/abs/1302.1525>
- Cont, R.: Model uncertainty and its impact on the pricing of derivative instruments. *Mathematical Finance* **16**(3), 519–547 (2006) <https://doi.org/10.1111/j.1467-9965.2006.00281.x>
- Dey, D., Liu, T.Y., Sofman, B., Bagnell, J.A.: Efficient optimization of control libraries.

- Proceedings of the AAAI Conference on Artificial Intelligence **26**(1), 1983–1989 (2012) <https://doi.org/10.1609/aaai.v26i1.8383>
- Dolan, E.D., Moré, J.J.: Benchmarking optimization software with performance profiles. *Mathematical Programming* **91**(2), 201–213 (2002) <https://doi.org/10.1007/s101070100263>
- Dai, M., Zhang, Q., Zhu, Q.J.: Trend following trading under a regime switching model. *SIAM Journal on Financial Mathematics* **1**(1), 780–810 (2010) <https://doi.org/10.1137/090770552>
- Givan, R., Dean, T., Greig, M.: Equivalence notions and model minimization in Markov decision processes. *Artificial Intelligence* **147**(1–2), 163–223 (2003) [https://doi.org/10.1016/S0004-3702\(02\)00376-4](https://doi.org/10.1016/S0004-3702(02)00376-4)
- Gleixner, A., Hendel, G., Gamrath, G., Achterberg, T., Bastubbe, M., Berthold, T., Christophel, P.M., Jarck, K., Koch, T., Linderoth, J., Lübbecke, M., Mittelman, H.D., Ozyurt, D., Ralphs, T.K., Salvagnin, D., Shinano, Y.: MIPLIB 2017: Data-driven compilation of the 6th mixed-integer programming library. *Mathematical Programming Computation* (2021) <https://doi.org/10.1007/s12532-020-00194-3>
- Golovin, D., Krause, A.: Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *Journal of Artificial Intelligence Research* **42**, 427–486 (2011) <https://doi.org/10.1613/jair.3278>
- Glasserman, P., Xu, X.: Robust risk measurement and model risk. *Quantitative Finance* **14**(1), 29–58 (2014) <https://doi.org/10.1080/14697688.2013.822989>
- Harvey, C.R., Liu, Y., Zhu, H.: ... and the cross-section of expected returns. *The Review of Financial Studies* **29**(1), 5–68 (2016) <https://doi.org/10.1093/rfs/hhv059>
- Karp, R.M.: Reducibility among combinatorial problems. In: Miller, R.E., Thatcher, J.W., Bohlinger, J.D. (eds.) *Complexity of Computer Computations*, pp. 85–103. Springer, Boston, MA (1972). https://doi.org/10.1007/978-1-4684-2001-2_9. https://link.springer.com/chapter/10.1007/978-1-4684-2001-2_9
- Lakner, P.: Optimal trading strategy for an investor: The case of partial information. *Stochastic Processes and their Applications* **76**(1), 77–97 (1998) [https://doi.org/10.1016/S0304-4149\(98\)00032-5](https://doi.org/10.1016/S0304-4149(98)00032-5)
- Li, K., Nyström, K., Olofsson, M.: Optimal switching problems under partial information. *Monte Carlo Methods and Applications* **21**(2), 91–120 (2015) <https://doi.org/10.1515/mcma-2014-0013>
- Mutti, M., Del Col, S., Restelli, M.: Reward-free policy space compression

- for reinforcement learning. In: Proceedings of the 25th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 151, pp. 3187–3203. PMLR, A Virtual Conference (2022). <https://proceedings.mlr.press/v151/mutti22a.html>
- McLean, R.D., Pontiff, J.: Does academic research destroy stock return predictability? *The Journal of Finance* **71**(1), 5–32 (2016) <https://doi.org/10.1111/jofi.12365>
- Moody, J.E., Saffell, M.: Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks* **12**(4), 875–889 (2001) <https://doi.org/10.1109/72.935097>
- Nemhauser, G.L., Wolsey, L.A., Fisher, M.L.: An analysis of approximations for maximizing submodular set functions—i. *Mathematical Programming* **14**, 265–294 (1978) <https://doi.org/10.1007/BF01588971>
- Pan, C., Ren, L., Xiong, L., Li, Y., Wei, W., Yang, X.: Regime-adaptive continual learning for portfolio management. In: Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2026). Accepted for publication; arXiv:2606.00143. <https://arxiv.org/abs/2606.00143>
- Russell, S.J., Wefald, E.: Principles of metareasoning. *Artificial Intelligence* **49**(1–3), 361–395 (1991) [https://doi.org/10.1016/0004-3702\(91\)90015-C](https://doi.org/10.1016/0004-3702(91)90015-C)
- Sondik, E.J.: The optimal control of partially observable markov processes over the infinite horizon: Discounted costs. *Operations Research* **26**(2), 282–304 (1978) <https://doi.org/10.1287/opre.26.2.282>
- Smallwood, R.D., Sondik, E.J.: The optimal control of partially observable markov processes over a finite horizon. *Operations Research* **21**(5), 1071–1088 (1973) <https://doi.org/10.1287/opre.21.5.1071>
- Sullivan, R., Timmermann, A., White, H.: Data-snooping, technical trading rule performance, and the bootstrap. *The Journal of Finance* **54**(5), 1647–1691 (1999) <https://doi.org/10.1111/0022-1082.00163>
- Weitzman, M.L.: Optimal search for the best alternative. *Econometrica* **47**(3), 641–654 (1979) <https://doi.org/10.2307/1910412>
- White, C.C.: Procedures for the solution of a finite-horizon, partially observed, semi-markov optimization problem. *Operations Research* **24**(2), 348–358 (1976) <https://doi.org/10.1287/opre.24.2.348>
- White, H.: A reality check for data snooping. *Econometrica* **68**(5), 1097–1126 (2000) <https://doi.org/10.1111/1468-0262.00152>
- Wolsey, L.A.: An analysis of the greedy algorithm for the submodular set covering problem. *Combinatorica* **2**, 385–393 (1982) <https://doi.org/10.1007/BF02579435>